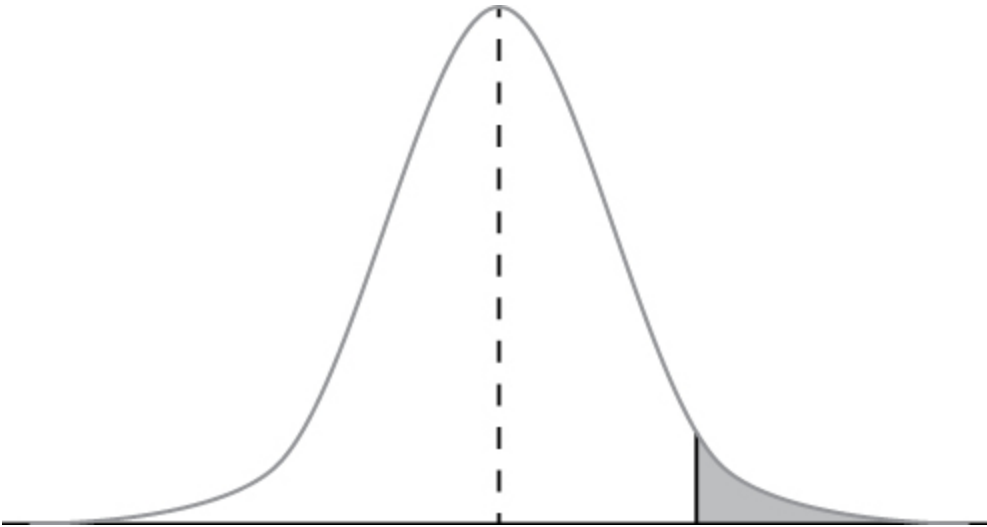




[Figure 5-3](#): A bell curve, or Gaussian normal curve (source: Wikimedia Commons)

The bell curve, like the golden ratio, appears in many sometimes surprising places in math and the universe. In this case, we interpret the close resemblance between our overlapping sums test results and the bell curve as evidence that our PRNG resembles true randomness.

Knowledge of the deep mathematics of randomness can help you as you design random number generators. However, you can do almost as well just by sticking with a commonsense idea of how to win at roulette.

Linear Feedback Shift Registers

LCGs are easy to implement but are not sophisticated enough for many applications of PRNGs; a savvy roulette player could crack an LCG in no time at all. Let's look at a more advanced and reliable type of algorithm called *linear feedback shift registers* (LFSRs), which can serve as a jumping-off point for the advanced study of PRNG algorithms.

LFSRs were designed with computer architecture in mind. At the lowest level, data in computers is stored as a series of 0s and 1s called *bits*. We can illustrate a potential string of 10 bits as shown in [Figure 5-4](#).